

Social Networks Analysis and the Law: A Proof of Concept for Argentina

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Abstract

Social network analysis (SNA) is a methodology to study complex systems. It has been used for many purposes: from studying the structure of English trade in the East Indies between 1601 and 1833 to the temporal structure of consensus formation in science. While its history goes back to the early twentieth century, its expansion follows the availability of computers as a way of gathering, organizing and transforming the data necessary to construct a “social network”. This paper argues that SNA is a powerful tool for studying legal systems, reviews some of the literature produced so far using it and argues that it can shed relevant insights into important legal questions. It describes the conditions necessary for successful SNA research and discusses possible ways of overcoming challenges in developing countries. It presents four preliminary approaches on Argentina: on the functioning of legislative committees in the House of Representatives, on the patterns of precedent citations in Supreme Court decisions, on the structure of the legal field in the city of Buenos Aires and on the structure of a legal advocacy clique behind a successful public interest litigation campaign.

Social networks analysis (SNA) is a social inquiry method that is useful for studying complex systems and the relationships that exist between the different actors that form them. It has been used for many purposes: from studying the structure of English trade in the East Indies between 1601 and 1833 (Erikson and Bearman 2006), the temporal structure of consensus formation in science (Shwed and Bearman 2010) and the dynamics of information flows in electoral campaigns (Benkler, Faris, and Roberts 2018). While its history goes back to the early twentieth century, its expansion follows the availability of computers as a way of gathering, organizing and transforming the data necessary to construct a “social network”. This paper argues that SNA is a

powerful tool for studying legal systems, and proceeds in the following way. In the first part, I overview basic concepts of SNA, draw a brief history of the field and review a few of the studies that have used it to research legal institutions. The second section discusses the necessary conditions for SNA to be deployed effectively, which include access to adequate data. This poses a challenge for research in developing countries, where data is often not available, it is available in formats that prevent automatic processing, it is incomplete or its authenticity is questioned. In the third section I argue—taking Argentina as a case study—that those challenges can be met. I do so by exploring four research ideas, one developed in the course of a technical advisory report and three in draft form: on the functioning of legislative committees in the House of Representatives, on the patterns of precedent citations in Supreme Court decisions, on the structure of the legal field in the city of Buenos Aires and on the structure of a legal advocacy clique behind a successful public interest litigation campaign.

I A brief history of Social Networks Analysis

The general consensus is that SNA was born in the 1930s, thanks—mainly—to the work of Jacob Moreno, a Romanian emigrate to the United States with background training in psychiatry and eclectic academic interests. In his 1934 *Who Shall Survive? A new Approach to the Problem of Human Interrelations*, Moreno used quantitative methods to describe and assess human relations, and used graphs and graph theory to describe them (Moreno 1953). Moreno's book "was a signal event in the history of social networks analysis ... a turning point for the development of the field" (Freeman 2004, 7). Moreno described what then was called *sociometry* as the use of quantitative methods to "inquire into the evolution and organization of groups and the position of individuals within them."¹ These methods relied heavily on the systematic gathering

¹ Moreno (1953), 11. His work explored behaviors and friendship networks in school children and was influenced by Helen Jennings, a psychologist specialized in empirical research working at Columbia University. For Freeman, Jennings's influence on Moreno was "immense" for he "had no research training at all, and until this point, he had shown absolutely no interest in systematic research. All of his earlier publications were couched in heavy mysticism. But his newer works—those produced during his collaboration with Jennings—were comparatively systematic and were, for the first time, empirically grounded". The obvious conclusion is that, though the intuitive ideas came from Moreno, the completed research and the publications drew heavily on the contributions of Jennings ... was not only a collaborator but she was, in Moreno's terms, a very powerful 'muse'. See Freeman (2004), 36. On the other hand, it should be noted that earlier works had looked at the same phenomenon with similar techniques. For instance, Almack study on friends networks who asked children to name who they would invite to a party (See Almack (1922)) and in the other observing who played

of data, the use of graphs to represent the structures of the networks that the data revealed, and the use of mathematical formulas to test and describe those structures.²

SNA as a field, however, failed to materialize at the moment, for various reasons (See Freeman 2004, chaps. 5–7 (discussing the so-called work produced during the “Dark Ages”, in the 1940s-1960s)). It was only in the 1970s that something of a renaissance happened in Harvard, under the leadership of Harrison Coyler White (Freeman 2004, 129–32). But it was the expansion of computers what really bolstered the popularity of the method (Freeman 2004, 139 (“Computers have played a key role in the development of social networks analysis...”), as part of the broader trend on computational social sciences.³ There is a reason for that: the complex mathematical formulas that can assist the researcher in assessing the structure of a given network can be easily coded in software and then *ran* against a dataset containing the necessary information. As Freeman puts it, “by providing a standard approach to data analysis, these programs played a role in the development of the social network research community” (Freeman 2004, 141). In that sense, these computer programs help researchers in three of the core features of SNA identified by Freeman: the systematic gathering of empirical data, the use of graphic imagery to represent the networks being researched, and the use of mathematical and computational models to analyze the relevant structures (Freeman 2004, 3 (the fourth is a “structural intuition” that leads the researched into studying the ties linking different social actors)).

Now, what are the main *insights* SNA as a methodology of social research could reveal regarding its object of study? Consider, for instance, the random networks produced in Figure 1.

These are two social networks, in which each node represents a person, and each edge (the links between the different nodes) represents a connection between the two, which could be e.g. “acquaintance.”⁴ Figure 1.a shows a totally

with whom on period of free play (See Wellman (1926)).

² This was a later development that Freeman attributes to Charles Lazarsfeld, a mathematic sociologist working in Columbia University. See Freeman (2004), 38

³ On this link, see Lettieri and Faro (2012)

⁴ Caiani (2014), 371 (“...the ties that link nodes can represent any sort of relations between them ... relations of cooperation, disagreement, or influence. Broadly speaking, the types of social relations that can be represented through network data include a) social roles (e.g., brother; father; boss; teacher); b) affective relations (e.g., friendship; positive or negative evaluations, such as like, respect, hate, etc.); c) cognitive relations (e.g., knowledge; information); d) organizational relations (co-organization; exchange of information; transfer of resources; co-occurrence, for instance being in the same political party; co-participating in a demonstration, etc.); e) actions (e.g., talking; having lunch...) ... and so on”.

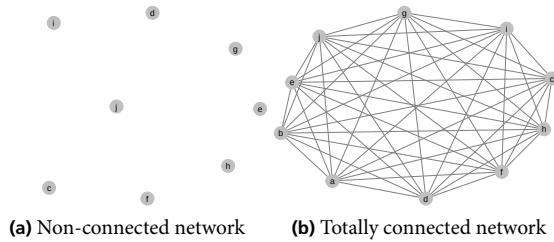


Figure 1: Networks Examples

disconnected network, while Figure 1.b shows one that is fully integrated, in which each node is connected to every single node in the network. Most real networks fall somewhere in the middle of both possibilities.

Generally speaking, network information can be gathered in the form of a matrix, specially if networks are not particularly large (Table 1). Connections between actors are represented by ones, while their absence is represented by zeros.

Table 1: Matrix of Random Network 1

	a	b	c	d	e	f	g	h	i	j
a	0	0	1	0	0	0	1	0	0	1
b	0	0	1	0	0	1	1	1	1	1
c	1	1	0	0	0	0	0	1	1	1
d	0	0	0	0	0	0	1	1	0	1
e	0	0	0	0	0	0	0	0	0	1
f	0	1	0	0	0	0	0	1	1	0
g	1	1	0	1	0	0	0	0	1	1
h	0	1	1	1	0	1	0	0	0	1
i	0	1	1	0	0	1	1	0	0	1
j	1	1	1	1	1	0	1	1	1	0

Nevertheless, the graphic representation of a social network is generally a useful way of seeing the data from a different perspective. Consider, for instance, Figure 2. E seems to be quite removed from the network, her role perhaps is not important or—perhaps—is structurally different: e.g., E may be a supervisor of a team of researchers, and only has a relationship with the team leader J who is in charge of reporting. These kinds of observations could be theoretically informed by e.g. what we know about how research labs work and, from that perspective, graphs can assist us in better explaining e.g. work dynamics, personal influence, collaboration dynamics, and so on.

What graphs can tell us in this way is illustrative and descriptive: graphs help us to see the world from a different standpoint, but by themselves do very

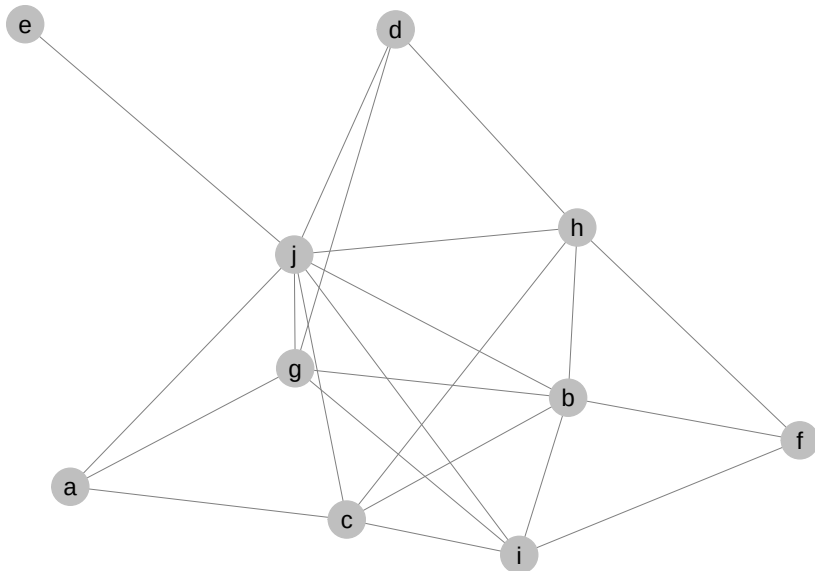


Figure 2: Random Network 1

for instance, for studying collaboration patterns between two different groups of researchers.

But SNA is much more than that: it gives us ways of mathematically studying a given network to uncover features that scape the eye. In that way, we can assess the structural features of a network and—on occasions—those parameters may help us see what we otherwise would not see by simply assessing the data through theory or purely qualitative methods (e.g., by producing a survey on all the members, in-depth interviews, and so on).

One of the more fundamental parameters of networks are their *density*, which measures the amount of total edges in a network against the amount of total possible edges of a network. In Figure 2, the density is e.g. 0.23, falling between the middle of a totally disconnected network equal to 0 (Figure 1.a) and a fully integrated network equal to 1 (Figure 1.b). Density can be a useful tool to compare networks of the same size, but when networks are of a different size (e.g., one lab has more members than others) its usefulness decreases (Scott 2011, 87 (“The crucial problem with the existing measures of density is that density is relative to size, which makes it difficult to use the measure in cmparisions of radically different sized graphs”). For an example of a useful

comparison of similar-sized networks, see @caiani2014, 383-385 (comparing networks of far-right organizations in Germany, Italy and the United States)). It played a key role in one of the pioneering studies on urban relations, the study produced by Barry Wellman in East York in the late 1960s (Wellman 1979).

Another major concept usually analyzed in social networks is *degree centrality*, that is, the amount of links or edges that a given node of a network has. The more it has, the more “central” the node is. From this measure it can be inferred that central nodes play important functions within the network. Centrality should be distinguished between *indegree* and *outdegree* in directed graphs such as e.g. the one presented in Figure 3, taken from Yang, Keller and Zheng (Yang, Keller, and Zheng 2016, 59)

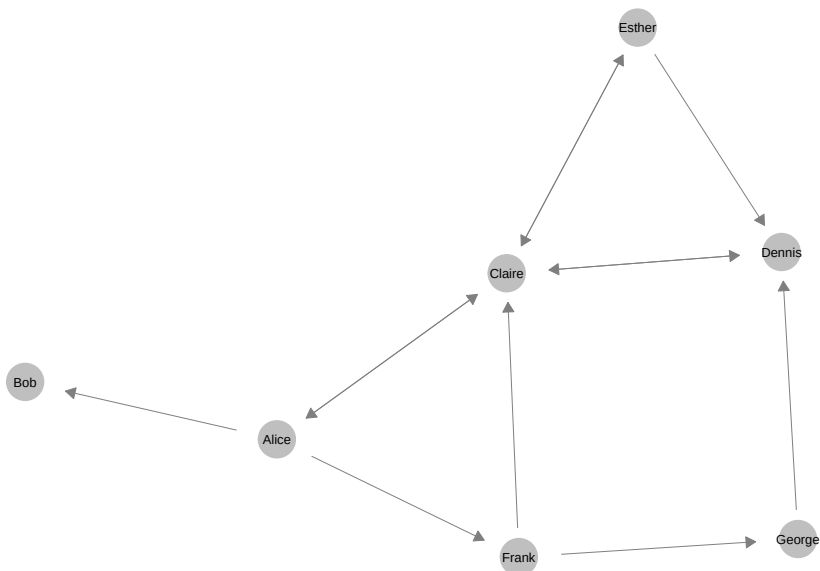


Figure 3: The Yang Friendship Network

In the Yang friendship network, Claire is the node with highest *in-degree centrality* (0.6) and she shares that position when considering *out-degree centrality* with Alice (0.5 each). Depending on the way the data was gathered and the kind of social structures the network is representing, high centrality can reveal important roles within the network. Consider, e.g., two examples: the pioneering work of Moreno studying school children identified pupils with high indegree values as the most popular within the network (Moreno 1953, 25

(“...others attracted so many choices that they captured the center of the stage like stars”). On the other hand, in-degree centrality is a fundamental part of Google’s proprietary algorithm PageRank®, which is a fundamental way of measuring the “importance” of websites *vis à vis* users’ queries and is—thus—a central building block of the world wide web (See e.g. Fortunato et al. 2008 (arguing that in-degree centrality could be used to approximate PageRank®)).

Another related value is *betweenness centrality*, which “measures the extent to which a particular point lies ‘between’ the various other points in the graph” (Scott 2011, 99). Generally, high in-degree centrality may yield high betweenness centrality, but on occasions betweenness centrality may reveal that a not particularly central node within the network may play important brokerage or gate-keeping roles (Scott 2011, 99). As Gould and Fernández put it, “actors in monopoly positions of moderate centrality are at least as powerful as actors in central positions” (Gould and Fernandez 1989, 90). In that sense, if the network represented in Figure 2 shows betweenness scores that follow in-degree and out-degree centrality scores (Alice = 0.4; Claire = 0.6), in a network with a different structure, brokerage or gate-keeping functions become clearer as in Figure 4, where node 4 plays that role and its betweenness values reveal it (0.53, followed by 7 with 0.25). In the latter case, node 4 is also a *cut-point*, a node which removal would create “two disconnected components” (Scott 2011, 118).

Another feature of networks that is usually useful to look at are *subgraphs*, that is, those groups of nodes that share “some defining feature drawn from the mathematical principles of graph theory: the connectedness of its points, the intensity of their connection, and so on” (Scott 2011, 115). When looking at these different “parts” of networks some features are generally useful from the point of view an inquiry. For instance, *cycles* in directed networks: they are paths “that return to its own starting point”, e.g., the path that goes Frank → George → Dennis → Claire → Alice → Frank in Figure 3. Finding paths in large networks can be useful for theoretical reasons: often, it is not unwarranted to assume that “the direction of a line” indicate “the possible flow of some resource or facility, such as money, power, or information.”⁵

On the other hand, components of networks can be studied in terms of how different nodes connect to one another, specially searching for those “especially cohesive or intensely connected point” that form the graph “center” (Scott 2011, 122). In that sense, *k-cores* use the degrees of the points as a measure of intensity, while *m-cores* rely on the multiplicities of lines, and are gen-

⁵ Scott (2011), 117 (but, note, “it can be difficult to give a substantive sociological interpretation to long paths of connection”, p. 120).

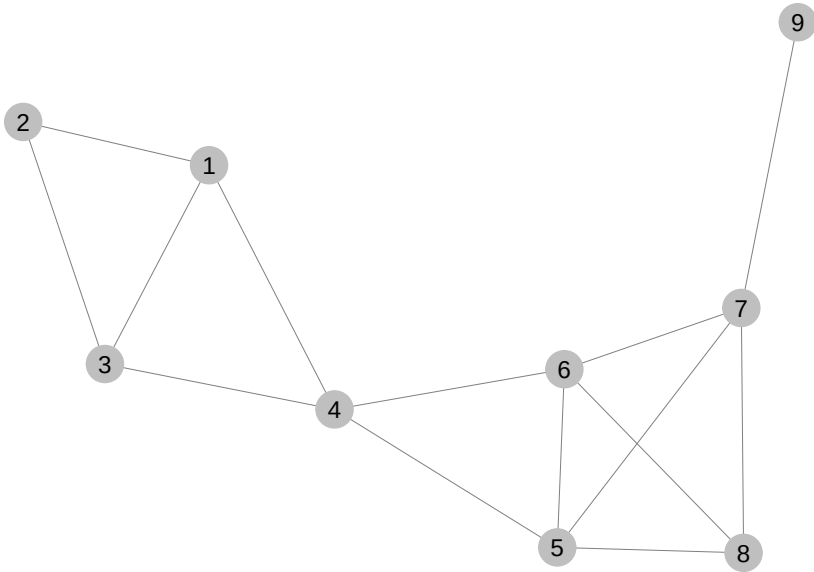


Figure 4: Random Network 2

erally useful shortcuts to find important nodes in large networks. On the other hand, another way of studying subgraphs is by looking at the different *cliques* that form it, defined as “the maximal subsets of points in which each point is in a direct and reciprocal relation with all others” (Scott 2011, 127). In directed graphs, such as Figure 2, if only reciprocal relations are considered the only cliques within the network would be the three that form around Claire, with Alice, Esther and Dennis. Scott claims that *n-cliques* are much closer to “people’s everyday understanding” of the word “clique”, and *n* stands for the “maximum path length at which members of the clique will be regarded as connected” (Scott 2011, 127). Similarly, *clusters* are sets of points “that are judged to be similar with respect to some relational property” as e.g. legislative committees that share a lot of members.

These are just a handful of the concepts SNA uses to study social networks; their usefulness depends on the kind of structure being studied and the theoretical assumptions of researchers. In that sense, and as all sorts of methodological approaches, SNA is deeply affected by the theoretical frame through which questions are posed and answers are sought. But it is a very flexible methodology that can and has been used to study different things. Here I would like to point towards a few studies that have used SNA to study legal institutions.

For instance, Fowler et al. have used SNA to study citation patterns of courts in the United States (Fowler et al. 2006; Fowler and Jeon 2008), following—within the legal field—the tradition of bibliographic studies. Others have expanded this approach in different directions (Geist 2009; Robinson 2010; Whalen 2013; Black and Spriggs 2013) and looked at different jurisdictions, such as Canada (Neale 2013), the European Union (Jacob 2014; Koniaris, Anagnostopoulos, and Vassiliou 2014), Italy (Agnoloni and Pagallo 2015), India (Minocha, Singh, and Srivastava 2015) and the Netherlands (Wyner and Casini 2017), to name a few. Research on legislatures has also used SNA to study e.g. the way committees are formed and distributed among parties, for instance in the House of Representatives of the United States (Porter et al. 2005) or how legislation co-sponsorship works in Perú (Lee, Magallanes, and Porter 2017). Some have used SNA to measure the extent to which clerks influence judicial opinions in the United States Supreme Court (Bonica et al. 2019) and how different States' attorney generals form coalitions around *amicus* campaigns (Gleason 2018). Others have used SNA to rank lawyers based on the relative importance of cases in Brazil (Ribeiro and Figueiredo 2017). Generally speaking, all sorts of empirical questions about the law can be addressed through SNA, for the legal system is complex and different parts of it can be understood in relational terms (Emirbayer 1997), such as e.g., the relationships that exist between cases, between judges and other public officials, between lawyers, between statutes, and so on.

II SNA and the Legal Field: Challenges and Opportunities

The use of SNA for legal inquiry seems to form a growing body of scholarship that is both recent and modest. My take is that SNA is useful when it is based on sound theoretical foundations: without them, qualitative analyses of the law always run the risk of fixating on the minutiae of data of secondary importance. When SNA works it does so as a *tool* — when it is used to perform certain task, in this case of an analytical nature. What are the conditions under which these analyses can benefit legal research? I will approach this question from the standpoint of research in developing countries where the kind of information that is necessary for SNA is not easily available: where e.g. open data policies are poorly implemented or non-existent, where legal research services are not particularly sophisticated, and where the legal academy is not a full-time, well-funded endeavor. Argentina is one of those countries many countries in Latin America more or less fit this description. Hence, it makes sense to keep in mind whether the conditions necessary for SNA can be met in such contexts.

The first condition for successful SNA research seems to be access to the

necessary data to construct social networks. This data is often difficult to come by and researchers on the legal field are specially ill positioned to gather it, because of methodological short-comings, insufficient full-time positions on the the professors' corps, and so on. SNA suffers in that sense from a kind of affliction that affects all empirical and socio-legal research. You have to put in the time to produce innovative scholarship. If a full-time job is not available, as it is often the case in countries such as these, scholarship becomes a part-time thing and other approaches become more appealing. SNA research seems to be particularly time-consuming when studying actors within the legal field and their relationships, for the data for SNA research is usually gathered through questionnaires and interviews (Wasserman and Faust 1994, 45). But advances in computer science makes data-gathering less onerous and more accessible, and these generally follows the kind of research seen until now, such as the study of citation patterns in legal cases (Fowler et al. 2006), affiliation networks in different organizations (e.g. law firms or the judiciary) (Katz and Stafford 2010), the structure of the legal academy (Katz et al. 2011), and so on.

The challenge however is that this information is often not in a format that can be easily gathered, manipulated and organized in the formats necessary to build a social network. Consider, for instance, the following example: most of the research on citation networks is based on access to proprietary databases that have gathered the cases and mapped the citations already.⁶ This is a starting point that is usually not available in developing countries, for legal research services are much more rudimentary and limited. The problem—as I will show in the next section—can be easily (or not too hardly) addressed through a combination of text-mining strategies and web scrapping techniques.⁷

Access to data that is useful for building social networks is also improving: open data policies in Latin America seem to be moving in the right direction and several datasets that seem useful for SNA research are already available, such as e.g. the dataset on judicial officers at the federal level in Argentina.⁸ On the other hand, web scrapping techniques can fill the gaps in lots of cases. However, for some research project, old-school, time-consuming social sci-

⁶ See e.g. Fowler et al. (2006), 327 (“We began with the list of all per curiam, signed, and in-chambers’ opinions published in the United States Reports provided on the U.S. Supreme Court’s Web site through its ‘Case Citation Finder’ link ... We did so by first comparing the Court’s list with the one we compiled through a series of Lexis-Nexis searches using the U.S. Supreme Court Cases library”); Robinson (2010); Clark and Lauderdale (2012)

⁷ Or, as Lettieri and Faro put it, “automated information extraction”. See Lettieri and Faro (2012)

⁸ Ramos (2017), 5 (“... it is possible to observe that East Asia & Pacific and Latin America & Caribbean had a big improvement in readiness and implementation especially in comparison to Europe”); Datos.gob.ar (2020)

ences techniques such as interviews, surveys and questionnaires are still necessary.

A second condition of SNA research on the legal field is obvious but should be mentioned: sound theoretical perspectives to produce adequate questions. It is only from a sound theory that SNA can be deployed effectively. To study citation patterns of judicial decisions we need to know what a precedent is and is supposed to be in a given legal system; to study how legislative committees work we need to know the function committees play in the legislative process and the function they are expected to play; to study the structure of the legal field we need to know how legal fields are usually organized and what are the dynamics that fuel different actors within it (Bourdieu 1987), and so on. Theories are necessary to give purpose to the inquiry, to identify the variables that are presumably useful to look at and—thus—to guide the empirical inquiry in ways that are useful to reject, support or modify the underlying theory.

A third condition seems to be particularly useful in this case: SNA for legal research needs a community of inquiry, that is, a handful of scholars that are interested in following the same methodological approach. It is noteworthy in that sense that lots of the works identified in the previous section have been produced through collaboration. SNA seems specially prone to this; different skills and backgrounds, when combined, can strengthen arguments and the analysis of complex data. A community of inquiry and co-authorship seems specially useful for the first stages of research: to test provisional hypothesis, refine theoretical arguments, and—specially—to divide the usually boring, time-consuming and prone-to-error process of building the datasets from where social networks can be built and analysis be deployed.

I will address these challenges in the course of the exploratory analyses that follow, but it is useful to keep them in mind as part of a broader invitation to explore this methodology for the purpose of legal inquiry.

III Potential use of SNA for the study of legal institutions in Argentina

In this section I present four possible social networks analyses. One was produced in the course of a technical report for *Fundación Directorio Legislativo*, as part of a collaboration project with the House of Representatives of the Argentinean Congress (Álvarez Ugarte 2019). The other three are first sketches of ongoing projects. The goal is to suggest possible ways forward and show their potential. In that sense, the last three subsections should be read as proofs of concept: to show that SNA is a viable methodology, that can be used for the

study of the Argentinean system and that it can shed light on questions that are timely and important, both to make and to answer.

1 *Studying Congressional Committees*

Porter et al. focused on the study of the affiliation networks of the committees and sub-committees of the House of Representatives of the United States Congress in order to find out how political parties “distribute” committee memberships and in order to identify clusters and measure the relative importance of the committees (Porter et al. 2005). In my study of the Argentinean system I sought to replicate that approach in the context of a technical report focused on finding more about how the committee system worked in order to suggest possible ways of improving the system. For that purpose, I produced two different analyses. One focused on interlocking dynamics within the rather large House committee system. Another took a sample of bills in order to see to what committees they were referred after they were introduced, and to follow the paths of bills that were referred to more than one committee (which I call *co-referrals*).

I expected that both approaches—the affiliation networks of committees and the way bills are distributed within the system—would shed light on the way committees work and help us confirm or refute hypotheses that are common in legislative studies: for instance, that committees produce *specialization* and that some committees are more important than others. (Fenno 1962; Asher 1974; Hamm, Hedlund, and Post 2011). In that sense, specialization has been proven in systems where legislative turnover is low, such as e.g. the United States (Matland and Studlar 2004). Does specialization occurs in systems where turnover is relatively high such as e.g. Argentina?⁹ My analysis suggests it does. Furthermore, the analysis on *co-referrals* of bills highlight the relative importance of certain committees and revealed some degree of overlapping in an already large committee structure, which is one of the largest in Latin America (Álvarez Ugarte 2019, 15).

Research was based on information publicly available in the House website, which was gathered through web scrapping techniques.¹⁰ A dataset of 17.224 bills introduced between 2016 and 2018 was produced.¹¹ This dataset

⁹ Santos (2006), 8 (Arguing that the rules of the system stimulate “stabler parliamentary careers. However, the electoral results show that in every election two out of three seats are occupied by freshmen”).

¹⁰ The code used to gather the data was written in Python by a data analyst working at Fundación Directorio Legislativo

¹¹ The dataset includes both legislative bills, as well as bills of a different nature, such as declarations and House resolutions

contained information on the date they were introduced and the committees where they had been referred. In case of *co-referrals*, an edgelist was produced to document the link between the committees that received the same piece of legislation. The data on committee membership was also produced by scraping the House website.

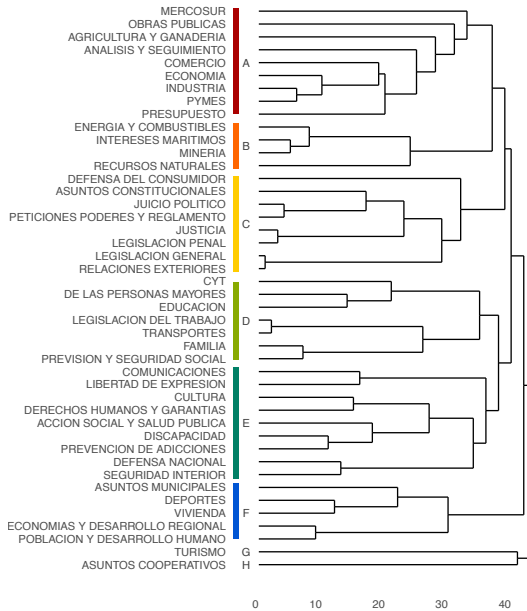


Figure 5: Dendrogram of House Committees

The data on membership reveals a clear pattern of specialization and dynamics of interlocking. As Figure 5 shows, the membership information organized in the form of a dendrogram show *clusters* with some degree of subject matter connections.

We can find those committees concerned with matters related to the economy (A), those involving natural resources (B) and those that appear to be more related to institutional matters (C). Then, two groups appear to be related to matters of rights (D and E), and then a miscellaneous cluster with more elusive connections (F). There are two strong outliers: the committees working on tourism (G) and cooperative businesses (G) do not share members with other committees. These sorts of analyses can reveal interesting patterns and obvious connections, such as e.g. the committees concerned with national defense and internal security, between mining and energy, between communication

policies and freedom of expression, and so on.

On the other hand, the information on *co-referrals* suggests that not all committees are equally important, as shown in Table 2 and Figure 6. When measured according to their *centrality*, the single most relevant committee in the House is the one in charge of discussing the budget (*Presupuesto y Hacienda*).

Similarly, the committees which seem to have the broader subject matter reach also score high under this light, such as *Legislación General* (General Legislation), *Asuntos Constitucionales* (Constitutional Matters), *Legislación del Trabajo* (Labor Legislation), and *Peticiones, Poderes y Reglamentos* (similar to the Rules Committee in the US House of Representatives). This kind of analysis allows to identify which committees perform essential functions and which are more peripheral. This conclusion is easy to explain and was confirmed through interviews with relevant officials: e.g. all the bills that presumably demand funds from the budget are automatically referred to the budget committee and the committee on General Legislation receives all the bills that do not clearly fit the jurisdiction of more specialized committees (Álvarez Ugarte 2019, 18–21).

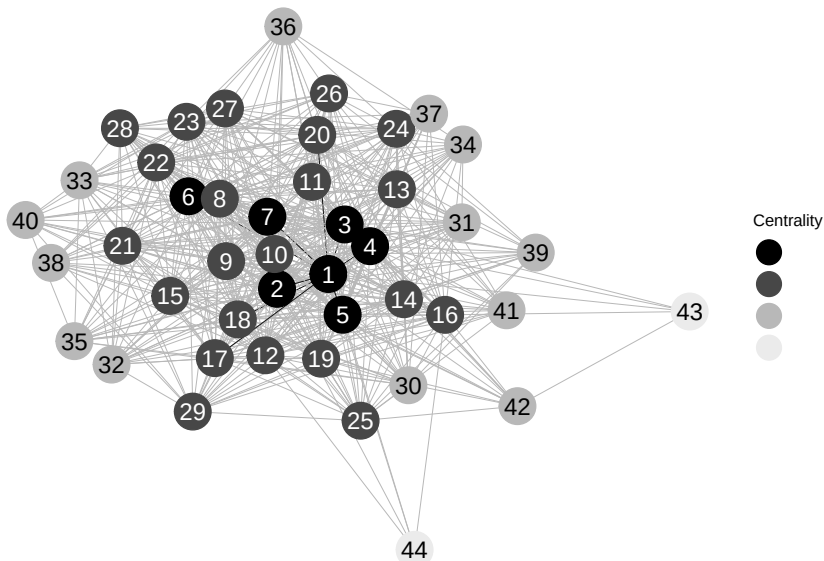


Figure 6: House Committees' Relationships Based on Co-Referrals

This approach could be expanded to further inquire into the legislative process. For instance, in Figure 7, an ego-network of the *Presupesto y Hacienda*

Table 2: Degree Centrality of Different House Committes

	id	Degree Centrality (Normalized)
Presupuesto y Hacienda	1	1.0000000
Legislacion General	2	0.9767442
Asuntos Constitucionales	3	0.9534884
Legislacion Del Trabajo	4	0.9302326
Peticiones Poderes y Reglamento	5	0.8837209
Accion Social y Salud Publica	6	0.8139535
Educacion	7	0.7906977
Comunicaciones	8	0.7441860
Justicia	9	0.7441860
Agricultura y Ganaderia	10	0.7209302
Legislacion Penal	11	0.7209302
Obras Publicas	12	0.6976744
Seguridad Interior	13	0.6976744
Transportes	14	0.6976744
Familia	15	0.6744186
Prevision y Seguridad Social	16	0.6744186
Relaciones Exteriores y Culto	17	0.6744186
Cultura	18	0.6511628
Discapacidad	19	0.6511628
Energia y Combustibles	20	0.6511628
Finanzas	21	0.6511628
Industria	22	0.6511628
Recursos Naturales	23	0.6511628
Comercio	24	0.6046512
Asuntos Cooperativos	25	0.5581395
Derechos Humanos y Garantias	26	0.5581395
Economias y Desarrollo Regional	27	0.5581395
Defensa Del Consumidor	28	0.5348837
Poblacion y Desarrollo Humano	29	0.5116279
Defensa Nacional	30	0.4883721
Deportes	31	0.4883721
De Las Personas Mayores	32	0.4651163
Pequeñas y Medianas Empresas	33	0.4651163
Prevencion de Adicciones	34	0.4418605
Ciencia	35	0.4186047
Economia	36	0.3953488
Intereses Maritimos	37	0.3720930
Mineria	38	0.3720930
Vivienda	39	0.3720930
Asuntos Municipales	40	0.3488372
Turismo	41	0.3488372
Libertad de Expresion	42	0.3023256
Analisis y Seguimiento	43	0.1162791
Mercosur	44	0.1162791

committee in cases of co-referrals—built with weighted edges—is a useful way of revealing what issues tend to demand more money than others. Similarly, interviews with House officials suggest that this sort of analysis would benefit from a longer study, spanning more years.¹²

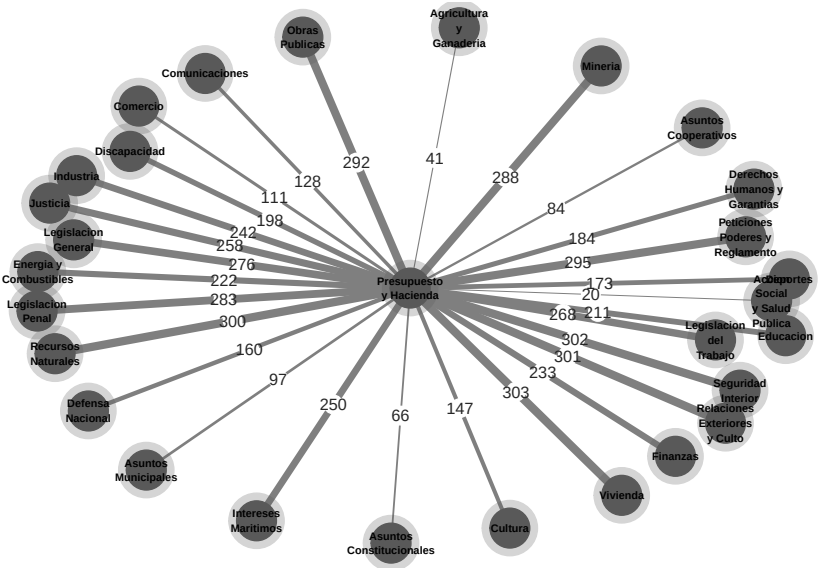


Figure 7: Weighted Edges of the Budget Committee

2 Citation Patterns of Supreme Court decisions

One of the most popular uses of SNA for legal research is to track case citation patterns of courts, in search of the most relevant or meaningful precedents. One of the earliest attempts to use SNA for this purpose was the paper that Fowler, Johnson, Spriggs, Jeon and Wahlbeck published in 2007 in *Political Analysis* (Fowler et al. 2006). They gathered 26,681 majority opinions by the Supreme Court decided between 1791 and 2005 and used the *Shepard's Citations* service currently owned by Lexis Nexis to gather the data on citations. This approach was later expanded by Fowler himself (Fowler and Jeon 2008),

¹² Indeed, one official interviewed suggested that committees gain or lose importance depending on the general political situation of the country; e.g., the committee that discusses labor legislation becomes more relevant when unemployment is high; the committee on public works generally does not have a lot of work but that changed when tariffs were discussed a few years ago, and so on

and was replicated by others in different contexts: to measure the importance of decisions by the Rehnquist Court (Robinson 2010), to study citation patterns around overruled cases (Whalen 2013) and to study other jurisdictions such as Canada, India, Italy, the European Court of Justice and the European Union (Neale 2013; Jacob 2014; Minocha, Singh, and Srivastava 2015; Koniari, Anagnostopoulos, and Vassiliou 2014; Agnoloni and Pagallo 2015).

The study of precedents through SNA has a few major advantages. First, it allows the researcher to study the importance of a case *at any point in time*: we can go back and see how important e.g. the Supreme Court of the United States considered *Marbury v. Madison* to be in the late nineteenth century (spoiler: not too important). Second, it allows for comparative research between different courts or jurisdictions. For instance, what is the status of the *pro life* decision by the Supreme Court of 2002 in *Portal de Belén* in provincial courts? While at the federal level the decision has been more or less ignored by the Court itself, which repudiated it in the *F. A. L.* decision of 2012, it is possible that in more conservative provinces the decision is still invoked as a relevant precedent. Third, and perhaps more importantly, these analyses can be produced with some degree of mathematical certainty.

Indeed, Fowler et al. developed a measure of *case importance* that takes into consideration not only the amount of times a case is cited, but the relative importance of the cases that cite it as well, following a similar approach than Google's PageRank® algorithm (Fowler et al. 2006, 330–32) and—specially—the work of Jon Kleinberg in *Authoritative Sources in a Hyperlinked Environment* (Kleinberg 1999). There, Kleinberg showed how the best way to measure the relative importance of a webpage for the purpose of identifying relevant results against search queries by Internet users was a combination of analyses of *in-degree* and *out-degree* links, the first because they revealed *authoritative* websites and the second because they revealed *hubs* of websites that pointed towards authoritative sources. The combination of both measures allows to identify websites that are important from the point of view of the query and the algorithm used to produce results showed, in testing, promising results.

Fowler et al. adapted Kleinberg's algorithm for studying Supreme Court decisions. They identified two measures of legal relevance of precedents: outwardly relevant cases (*hubs*), which are cases that cite other relevant cases and help “to define which decisions are pertinent to a given legal question” (Fowler et al. 2006, 330) and inwardly relevant cases (*authorities*), which are cases that are “widely cited by other prestigious decisions, meaning that judges see it as an integral part of the law” (Fowler and Jeon 2008, 330). Could their approach be replicated to study decisions by e.g. Argentina's Supreme Court? Fowler et

al. approach rested heavily on access to a research service such as *Shepard's Citations*; we meet, thus, a first difficulty: local law reports do not provide the same kind of detailed analysis of citation patterns, not even for Supreme Court decisions. The challenge, however, can be met through web-scraping techniques and a *rinse and repeat approach* that scraps the first set of decisions, gets the PDF files where decisions are, mine them to extract citations and build a data frame that grows with each iteration of the process (Figure 8).¹³

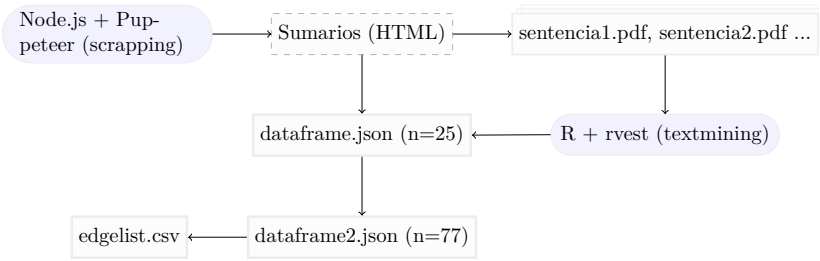


Figure 8: Scrapping / text mining workflow

Through this approach it is possible to gather the kind of data useful to build a citations network, which—as it is the cases with bibliographical studies, which in a way pioneered this approach—is always a *directed acyclic graph* (DAG): it always goes backwards (for cases can only cite previous decisions) and has no cycles, that is, is directed but not in a way that a path can return to the same vector of origin. Figure 9 presents a visualization of the network of decisions on freedom of expression (n=102) that was obtained through the *rinse and repeat* method described in Figure 8, colored according to the date of the decision and with the nodes sized according to the *authority score* produced by Kleinberg's algorithm -(Kleinberg 1999).

The approach yields interesting results. On the one hand, Kleinberg's *authority score* identifies the most widely cited cases, which include landmark decisions such as *Campillay* (n.46), *Costa* (n.47), *Granada* (n.67), *Patito* (n.8),

¹³ This approach produced solid results. While the scrapped citations of a case can be attached to the original data frame automatically, I did it manually in order to account for possible error in the scrapping script. I discussed this approach with Juan González Bertomeu. In discussion we reached—as I recall—a conclusion: automatic approaches are prone to error, but manual gathering is time-consuming. Probably, a mixed approach will yield better results in terms of the trade-offs involved between time and efficiency. On the other hand, complete networks are useful but not always necessary: depending on the research question involved, a time-limited or subject-matter scope may be as useful and may be built in shorter periods of time. The example I present here is limited by subject matter: 25 freedom of expression decisions, tagged as such by the Court itself, yielded citations to 77 additional decisions.

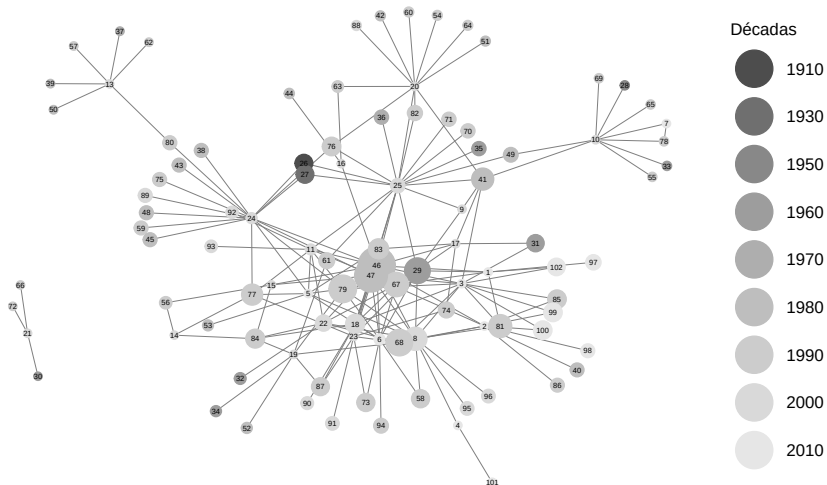


Figure 9: Freedom of Expression Cases (1914-2014)

Amarilla (n.81) and *Ponzetti de Balbin* (n.41). This generally follows what freedom of expression scholars identify as landmark decisions. Interestingly, the approach also reveals cases that one would not think as particularly central for the Supreme Court's jurisprudence on freedom of expression, such as *Pandolfi* (n.79), *Triacca* (n.68) and *Díaz* (n.83) or *Perini* (n.18). A first preliminary conclusion would suggest that those cases deserve a second look. Perhaps, they are used by the Court redundantly: e.g., they are non-landmark decisions that ratified previous landmark decisions and are cited nevertheless, to highlight the strength of the precedents being invoked. Perhaps, they are cited in specific contexts — e.g., in cases where the plaintiff is a public official or a judge, to invoke similarity. In any case, a careful scrutiny is warranted by this approach.

My own intuition is that Kleinberg's *authority* scores are more relevant than *hub* scores, at least in large networks,¹⁴ for they are prone to differ: cases that cite lots of other cases may or may not be particularly important for the field at hand for e.g. they may be concerned with other area of the law, something that happens e.g. when the Supreme Court had to deal with a complicated issue of standing before addressing the central question of the cases.¹⁵ Table 3 and Table 4 organize the top twenty decisions according to *authority* (Table 3) and

¹⁴ Note that Fowler et al., for explaining their methodological approach, use a 5-node network

¹⁵ Think, for instance, in the way the Supreme Court of Argentina dealt with the matter of the *amparo* in a decision about presidential decrees such as *Peralta*

hub scores (Table 4) and show this disparity.

Table 3: FOE cases organized according to Kleinberg's authority score.

Id	Year	Case	Hub score	Authority Score
46	1986	Campillay, Julio César	0.00	1.00
47	1987	Costa, Héctor Rubén	0.00	0.76
79	1997	Pandolfi, Oscar Raúl	0.00	0.52
68	1993	Triacca, Alberto Jorge	0.00	0.47
29	1960	Abal, Edelmiro,	0.00	0.44
67	1993	Granada, Jorge Horacio	0.00	0.39
8	2008	Patito Jose Angel	0.72	0.35
81	1998	Amarilla, Juan H	0.00	0.34
41	1984	Ponzetti de Balbín, Indalia	0.00	0.31
77	1996	Ramos, Juan José	0.00	0.27
83	1998	Díaz, Daniel Darío	0.00	0.24
18	2003	Perini, Carlos Alberto	0.58	0.22
84	1998	Kimel, Eduardo	0.00	0.21
76	1996	Monges, Analía M	0.00	0.20
100	2013	Canicoba Corral, Rodolfo	0.00	0.20
85	1998	Saucedo, Daniel Horacio	0.00	0.20
99	2012	Quantín, Norberto Julio	0.00	0.20
58	1991	Vago, Jorge Antonio	0.00	0.19
73	1996	Morales Solá, Joaquín Miguel	0.00	0.19
87	1998	Gómez, Jorge Miguel	0.00	0.18

Table 4: FOE cases organized according to Kleinberg's hub score.

Id	Year	Case	Hub score	Authority Score
3	2019	De Sanctis, Guillermo Horacio	1.00	0.00
6	2011	Melo Leopoldo Felipe	0.95	0.03
25	2001	Menem Carlos Saul	0.82	0.10
24	2001	Guazzoni, Carlos Alberto	0.78	0.03
2	2019	Martínez de Sucre, Virgilio Juan	0.77	0.00
8	2008	Patito Jose Angel	0.72	0.35
23	2002	Spacarstel, Néstor A.	0.72	0.00
18	2003	Perini, Carlos Alberto	0.58	0.22
1	2019	García, Stella Marys	0.53	0.00
11	2007	Sciammaro Liliana	0.52	0.03
17	2004	Roviralta, Huberto	0.46	0.00
22	2003	Burlando, Fernando Andrés	0.43	0.14
15	2004	González, Adriana Ruth	0.41	0.00
19	2003	Baquero Lazcano, Silvia	0.32	0.00
5	2013	Barrantes Juan Martin	0.31	0.00
16	2004	Vallejo, Guillermo Alberto	0.22	0.00
20	2003	R., S. J.	0.18	0.00
9	2007	Franco Julio Cesar	0.17	0.00

Id	Year	Case	Hub score	Authority Score
14	2006	Montejano, Ana Inés	0.11	0.00
10	2007	Editorial Rio Negro SA	0.10	0.00

A case that cites lots of other decisions and also scores high in terms of authority (is “cited” by other relevant decisions) may be an important part of the network and may deserve a careful scrutiny. Again, at this point this is speculative, but consider cases such as *Patito* (2008) and *Perini* (2003). two decisions that arguably sought to clarify the way different aspects of the case-law interacted with each other, namely the *Campillay* standard, the actual malice doctrine and relevance of the standing of the plaintiff for applying them. As Fowler et al. argue, *hubs* are likely cases that are grounded in the law, in the sense that provide a wide take on what the case-law say on the legal question at hand (Fowler et al. 2006, 330).

This approach, however, has more potential than simply helping us navigate complex citation networks to find insights that we would otherwise likely miss through pure doctrinal analyses. I would mention three. First, Fowler et al. argue that their method allows to track how cases become “more or less relevant over time, in comparison to other cases in the network, in a more valid way than extant measures” (Fowler et al. 2006, 344). Second, the approach can be used to “predict” which cases will be cited by lower courts, by going back and forth between Supreme Court cases and Appellate Court decisions. *Authority* and *hub* scores could reveal at which point a case is more likely to be cited by lower courts and at which point that would be unlikely, an outcome that depends on a more complex network but that seems to be important and worthy of development.¹⁶ This approach could also be used to study citation patterns of different courts as in González Bertomeu -(Gonzalez Bertomeu 2019) or to inquire into how respectful courts are of their own precedents, how judges use cases to argue for majority opinions or argue for dissents, and so on.

3 The Structure of the Legal Field in the City of Buenos Aires

Another interesting use of SNA would be to look at the structure of *the legal field* in a specific context. Indeed, it is well known that the legal field is formed by different players, acting in different positions of a highly hierarchical structure (Bourdieu 1987). Uncontroversial social theory suggests that this structure is important for the way the field operates: players exchange information, develop patterns of interaction and—generally speaking—deploy unequal so-

¹⁶ Fowler et al. (2006), 342-343 (discussing this approach)

cial capital for advancing their goals, which—as Bourdieu puts it—has to do with determining what the law is.¹⁷ SNA has been used to study these social structures. For instance, Katz and others have used it to study the structure of the federal judiciary (Katz and Stafford 2010) and of legal academia in the United States (Katz et al. 2011). In their first study, Katz et al. were able to identify central actors by studying the extent to which federal judges “placed” clerks in Supreme Court positions (Katz and Stafford 2010, 477–83), and used different SNA measures to classify the standing of judges, both from the Appellate and Supreme Court level (Katz and Stafford 2010, 496). In their second study, Katz and his colleagues studied the American professors’ corps based on a dataset built around education and hiring trends, which produced a directed network of the structure of legal academy (Katz et al. 2011). In this case, *out-degree* centrality appears to be the most relevant measure to look at: they were trying to find which Law Schools placed more professors in American Law Schools *at large* (Katz et al. 2011, 89–90).

In this case access to pre-processed information like the *Shepard’s Citations* will not do: these sorts of datasets must be built, with information carefully gathered and combined in a way that yields the kind of information with which a social network can be built. This seems possible in Argentina, even though information is spotty — at best. For instance, a rather complete dataset on the federal judiciary has been published by the Argentine government in the context of its open data policies of the last few years (Datos.gob.ar 2020, (n=1600)). It contains relevant information, including gender, the day in which the judge took office, and so on. But it is incomplete: in many cases it tracks positions rather than actual officers. Another dataset, this time of the *Defensoría General de la Nación*, shows the full potential of this policy: it includes the list of 2740 public officials, covering most of the personnel of this important part of the federal judiciary (OAIP 2020). Other information could be gathered: e.g., the law profession is regulated by law and the registry of active lawyers is public. It is not far fetched to either request or scrap the publicly available database. Finally, law schools should also be looked at. Curiously, this seems to be the information that appears to be least available: the Law School of the University of Buenos Aires—for instance—does not publish who teaches there. Some private law schools do provide this information with some level of detail but

¹⁷ Bourdieu (1987), 817 (“The juridical field is the site of a competition for monopoly of the right to determine the law. Within this field there occurs a confrontation among actors possessing a technical competence which is inevitably social and which consists essentially in the socially recognized capacity to interpret a corpus of texts sanctifying a correct or legitimized vision of the social world. It is essential to recognize this in order to take account both of the relative autonomy of the law and of the properly symbolic effect of ‘miscognition’ that results from the illusion of the law’s absolute autonomy in relation to external pressures”)

others do not.¹⁸

The kind of research question we may have define the kind of data we need to assess them empirically. For instance, if we would like to see the career paths that lead to the federal judiciary, the scrapping of personal resumés of judges would be the way to go. Another possibility is to gather the data on membership of different institutions in order to see the degree to which these are interlocked, an exercise which—in this case—is specially relevant to study Argentina’s legal professoriate, which is essentially part-time. This exercise will not find interlocking patterns between courts and law firms, for obvious reasons. But, if law-teaching is a part-time career, it is likely that university departments will show interlocking patterns between judges and lawyers, thus having the potential of being places of *brokerage* or *gatekeeping* (On this, see Wasserman and Faust 1994, 174; Gould and Fernandez 1989). This should have consequences, specially with regard to ethical concerns within the legal profession for potential conflicts of interest and influence peddling (Böhmer 2003, 2008) and—perhaps more crucially—in relation to the kind of legal academy, research and criticism that this kind of scholarship produces (Nino 1988; Böhmer 2014).

A tentative approach yields interesting results. By mixing web-scraping techniques and publicly available information it is possible to build, roughly and rather quickly, a dataset of judges, public prosecutors, public defendants, law professors and lawyers in top-law firms in the city of Buenos Aires (n=2594).¹⁹ By processing the information of public officials who belong to one or the other institution it is possible to build an *affiliation network*, that can be simplified by hiding people and showing only institutions connected to each other by the people they “share” (Figure 10). This seems to paint a rather telling picture of the structure of the legal field in the city of Buenos Aires, and highlights a fundamental feature: the extent to which courts are interlocked with the legal academy, in the form of top positions in the

¹⁸ E.g., it is easy to access the law professors at Universidad Di Tella, impossible in other prestigious private institutions such as the Universidad Católica Argentina or the Universidad Austral

¹⁹ The dataset should be considered a rough approximate to the issue, because it mixes data sources of various degrees of quality. The main reason for that is that to access a full list of professors at the Law School of the University of Buenos Aires is notoriously difficult. I managed to find, lost in an old version of the Law School website, a list of 781 professors (chairs and adjuncts), but it is not updated. On the contrary, it is a picture of what the Law School faculty looked like a few years ago. On the contrary, the dataset on judges is quite updated (June 2020). Presumably, data on law firms is up-to-date (it was scrapped from their websites in May 2020). In any case, the lack of updated information on the structure of the faculty of Buenos Aires main Law School makes this approach a provisional one, until—through a public information request—an updated version of that list can be gathered.

University of Buenos Aires Law School different departments.

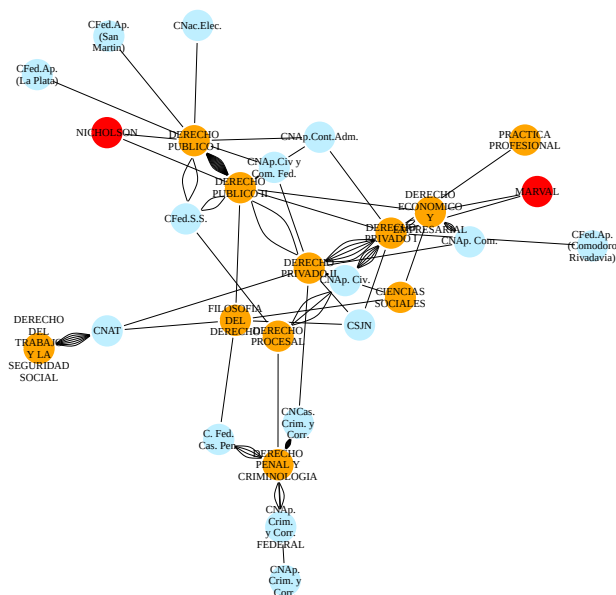


Figure 10: The structure of the legal field in the city of Buenos Aires

A careful scrutiny of this complex network reveals the patterns of the interlocking dynamics. Consider, for instance, the relationship between the High Court of Appeals on Labor with the department of Labor Law at the Law School. There, nine judges of the Court of Appeals hold adjunct positions at the Law School, seven at the Labor Law Department, one in Philosophy and one in Private Law. This subject matter connection repeats itself in other cases: Criminal Law judges hold positions at the Criminal Law Department, Civil Law judges hold positions at the Private Law Department, and Commercial Law judges hold positions at the Economic and Business Department (Figure 11).

The logic of interlocking is somewhat to be expected if legal research is not a full-time profession: somebody has to teach the law. The dataset used to build the social network of Figure 10 only scraps the surface of this feature of Argentina's legal field: it is focused only on those occupying the top positions of the different spheres in the city of Buenos Aires, and even though the amount of actors is significant, it is not even close to a full or semi-full picture. The information on law professors is particularly scarce: lots of actual teaching in

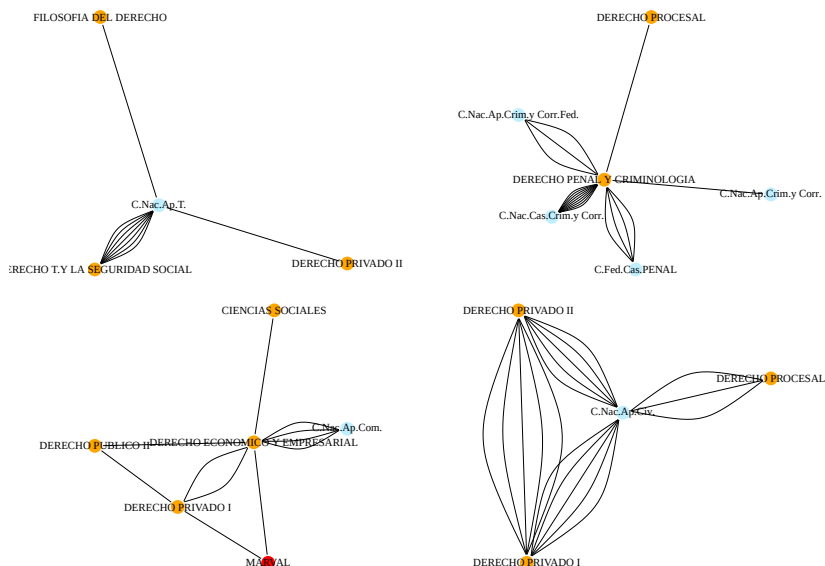


Figure 11: Interlocking dynamics through specialization (Four cases)

the system of chairs (*cátedras*) happen through persons who are not included in the sample, such as interim adjunct professors and teaching assistants of different degrees (JTPs, *ayudantes*, and so on). But the interlocking dynamic is important nevertheless. As Wasserman puts it,

“Common membership in organizations creates a linkage between people, and sharing members creates a linkage between groups that have one or more members in common. If we consider ties between actors or between events as potential conduits of information, then the connectedness of the affiliation network is important because information originating at any event (or with any actor) can potentially reach any other event (or other actor)” (Wasserman and Faust 1994, 317).

Information flow is, then, an important reason to study interlocking dynamics in the legal field, for they might play a significant role in the distribution of social and symbolic capital with which different positions of authority are gained within the field (Bourdieu 1987). These social links can be seen as “resources of individual actors that increase the probability of success of their own actions” (Caiani 2014, 370). This may have consequences in terms of how the field is structured, and there are reasons to believe that interlocking

is a problem the law as a field should address. But further research is needed, not only to expand the sample in a way that reaches more satisfactory boundaries of the network under scrutiny, but also to understand what this particular kind of interlocking means in practice. For instance, we know that the system of *cátedras* through which legal education is organized at the University of Buenos Aires is a space of socialization where ties between members are “more likely.”²⁰ What kind of socialization actually takes place there? What kind of career paths are aided by common membership to a *cátedra*? What kind of lateral and vertical moves happen within the system? What internal regulations—e.g., the system of interim professorships—impact the way interlocking works in practice? All these questions are begged by the structure of the network itself. Further quantitative and qualitative is needed to understand what interlocking means for the way Argentina’s legal field works in practice.

4 Advocacy Cliques in Public Interest Litigation Campaigns

Finally, SNA can be used to study social movements, and social movements are—arguably—a relevant actor to understand how the law is shaped (Post and Siegel 2013). These can be approached through SNA for they are formed by people and organizations in relation to one another, that exchange information, compete for resources and coordinate their actions (Tilly and Wood 2008). Social movements seeking legal change usually share certain features; social capital is organized in a certain way like e.g. around professionalized NGOs, or around specific professional roles, such as e.g. the activist or cause lawyer (Scheingold and Sarat 1998; Sarat and Scheingold 2006). They also engage in a very particular kind of relationship in which what flows is very specific: *demands* — requests for legal action, whether it is legislative reform, rights recognition or conflict adjudication (Scheingold [1974] 2004). Generally, these demands carry with them disputes about the meaning of the law (Cover 1983). Hence, SNA approaches can help us look at different aspects of social movement dynamics in relation to the law.

To follow the way social movements act demands something more than simply gathering data that is out there. Unlike citation patterns in judicial decisions, bill distributions and membership in a legislative committee system and affiliation networks within a social field, to study social movements demands

²⁰ Wasserman and Faust (1994), 296 (where Wasserman considers different possible interpretation of affiliation links. In particular, he reports that “some authors argue that individuals’ affiliations with events provide direct linkages between the actors and/or between the events”, a position I reject in favor of the position that argues that “contact among individuals who participate in the same social events provides conditions under which pairwise ties among individuals become more likely”)

to inquire into generally loose groupings of people and organizations, whose relationships with each other are hardly static: they are in permanent flux and are affected by the flux itself (Emirbayer 1997). These are dynamic entities that are difficult to grasp. In cases such as these, several challenges typical of social inquiry emerge forcefully, such as e.g. the problem of boundary specification (Emirbayer 1997, 303–5) and the definition of the “case” being studied (Ragin and Becker 1992). This is particular problematic for the extrapolation of consequences or the identification of causal effects. To illustrate some of these difficulties, but also the potential of the approach, consider the following example.

In the course of my doctoral dissertation, I studied the network of activists—mainly lawyers—behind the *F. A. L.* litigation campaign, how I called an effort to bring the Supreme Court of Argentina to adjudicate an abortion case that was in its docket and that was a likely candidate for dismissal under the famous mootness doctrine, for the abortion involved in the controversy had already happened. A first challenge in the approach was to identify the network, that is, the actors that were involved. Through a snow-sampling technique and departing from a handful of seed subjects I was able—at some point—to achieve theoretical exhaustion, when new subjects were simply not offering new insights or facts regarding the process that led to the *F. A. L.* decision and its aftermath. SNA was not particularly relevant for the story I was tracking. In particular, I was interested on “how constitutional claimants who succeed in eliciting public response secure official recognition of their views” (Siegel 2006, 1351). This meant tracing a process that I knew had occurred and finding out how it was carried out.

The snow-sampling approach allowed me to trace a process which was embedded in a broader effort to clarify a rather narrow issue of legal interpretation, and—in this particular case—asked the Court to do it as the highest court of the land. This effort meant creating normative arguments, engaging public officials with them, persuading them of making a decision and in a specific desired direction. It involved lobbying court officials and coordinating efforts and resources between different civil society organizations, legal academics and sympathetic public officials. SNA in itself is not of primary importance for this effort, but could it shed light on the network I described through archival research and—mainly—in depth personal interviews? In order to explore that possibility, I went back to the transcripts of the interviews conducted and identified the persons who had been mentioned as relevant for the purpose of the inquiry by the seed subjects and who were, then, part of the sample. This led to an unweighted matrix of 51 subjects who had been identified as relevant (Figure 12.a), a network that was later reduced by eliminating the vertices /

subjects who had been named only once.

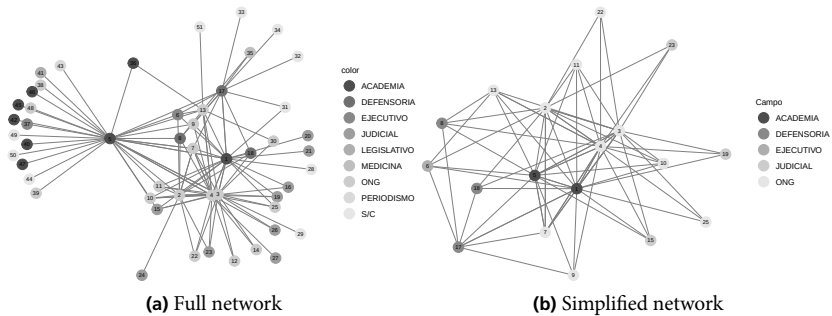


Figure 12: Legal Advocacy Clique Network

This is a rather standard network, in which edges are built based on whether they had been nominated by more than one subject in the course of the snowball sampling research. In itself is not particularly telling — exploring the structure of the network does not make much sense, for it is an obviously centralized network and subjects who had been interviewed will inherently appear more prominently than those who were not. But the links themselves could be made more complex, by adding weights according to some criteria which had to be coded in, such as the kind of reference made by the subject, whether they belonged at the time of the events to the same organization, and whether both subjects reported taking part in the same event. This led to seven possible ways in which an edge could be weighted.²¹ Weighted edges, represented in Figure 13, are more revealing of how this particular network of activists and public officials interacted in practice.

For instance, 1 and 5 (clique “A”) worked closely together and could be described as the main “creators” of the legal reform strategy that eventually succeeded in the *F. A. L.* case. On the other hand, 2:4 and 7 (clique “B”) worked in the same highly professionalized NGO that had recently decided to push

²¹ I added one point if any of the following conditions were met in the edge at hand: (a) whether the person was mentioned, a condition that verified in all cases; (b) whether the mention highlighted the importance of the nominated subject contribution; (c) whether the mention explicitly assigned a key role to the person being mentioned; (d) whether both subjects belonged to the same organization at the time of the events; (e) whether both subjects had belonged in the past to the same organization; (f) whether the data confirmed that both subjects had exchanged some kind of information; (g) whether both subjects participated in one of the events that had been identified as part of the litigation campaign. These categories are of course

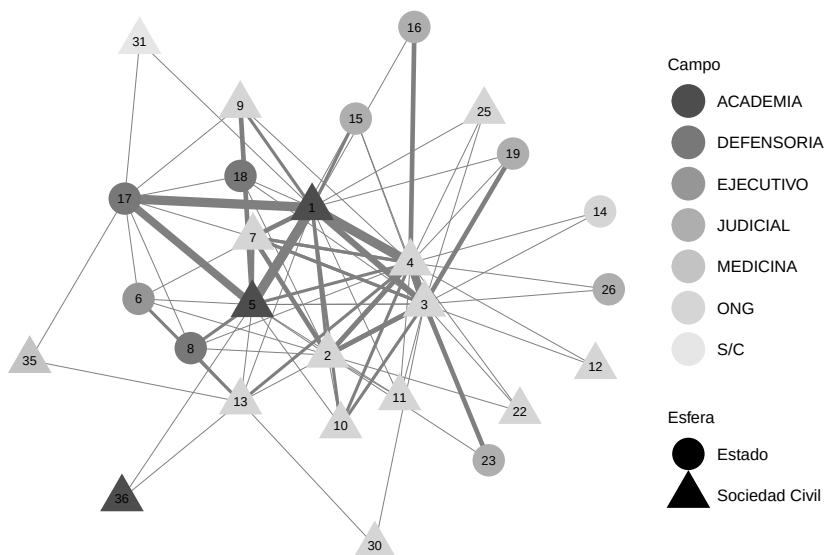


Figure 13: Legal Advocacy Clique Network (detailed)

forward an abortion agenda. These subjects appear to be at the center of the network which—at that point—sought to move forward through courts and, in particular, was looking for a case to reach the Supreme Court in order to solve a narrow legal question surrounded by profound uncertainty: how the exculpatory clauses of abortion prescribed in the Criminal Code since 1921 should be interpreted (whether narrowly or broadly) and whether these exculpatory clauses implied “a right” when their conditions had been met. A had come up with the idea and produced knowledge—arguments, reports, law review articles—in support of their position; B was working towards taking the abortion issue to court, and that had included—until then—an *amicus brief* strategy that had them present several *amicus curiae* briefs in front of courts in many provinces, where abortion cases had been heavily judicialized in the years before. At some point, A and B came together—literally, in a meeting which became very central during the process of interviews. It was in that meeting that the NGO received notice that one of the cases in which they presented an *amicus* brief had reached the Court. Several of the subjects in this central cliques played *brokerage* roles within the network, and the weighted nodes highlight it (Gould and Fernandez 1989). For instance, node 1 was an instructor in the province of Río Negro on gender and law, and a public de-

fendant that took part in the *F. A. L.* case was one of her students (node 17). Nodes 3 and 4, on the other hand, not only organized (with 2 and 7) an *amicus* strategy in front of the Supreme Court, but also asked—and got—meetings with some of the justices themselves (nodes 19, 23 and 26) or some of their aides (nodes 15 and 16).

In this case I have used SNA just to illustrate part of my research, in order to show how it can shed light on the area of legal inquiry I am particularly interested in (law and social movements). But I believe the approach could be used more extensively and expanded in at least the three following directions. First, it would be interesting to use SNA to study different cases, in which common criteria for boundaries are set. For instance, this particular litigation campaign could be compared to others. On the other hand, the codification of interview data could be made stronger by coding criteria before hand.²² There is lot to explore in that direction, in order to build robust networks that can aid in overcoming some of the perils of testimony as a source on empirical data.²³ Third, and finally, entire social movements can be looked at from the standpoint of SNA and different strategies can be deployed to build the necessary frameworks, some not as time-consuming as building them based on sampling methods out of in-depth personal interviews (For an example of that, *see* Caiani 2014). In any event, the method seems relevant for studying an underdeveloped area of legal inquiry, particularly in developing countries.

IV Conclusion

I have introduced social networks analysis as a methodology that is useful for legal research. I have presented its main concepts, scratching the surface of a growing body of scholarship that is used for multiple purposes. The review of the works that used SNA for studying legal system suggest different possible directions, most of them determined by sound theoretical concerns but—also—the availability of the necessary data. This should make us cautious for the potential of being in front of a problem that looks like a nail because we have a brand new hammer in our toolbox. Nevertheless, research so far is promising and scholars working in developing countries should consider this approach. So far, the insights produced through SNA analysis seem important, timely, and relevant. In the third section I sought to show how SNA for legal research is possible even when access to the necessary data is not ideal. For that purpose, the use of different strategies for getting, mining and transforming the

²² My own SNA approach was developed after interviews had been conducted, even though I organized since the beginning the information in the manner of a matrix

²³ On this, *see* Jerolmack and Khan (2014)

data seems essential, and makes SNA a subset of computational social studies (or, as Katz puts it, computational *legal* studies).²⁴ In any case, the usefulness of these tools makes this approach particularly prone to collaboration: lawyers usually lack the skills necessary to deploy this method effectively. Collaboration with sociologists, political scientists, economists and computer scientists seem necessary to exploit this approach in its full potential. On the other hand, as I mentioned in section 2, it seems important that some sort of community of inquiry grows around this approach. Otherwise, scholars doing SNA for legal research will end embedded in the many SNA networks around the world that seem to be growing. If so, SNA will still shed light on the facts of the world that are brought under its lens, but will not impact legal research, inquiry, and scholarship, in significant ways. There are reasons to suppose that it should.

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²⁴ See <https://computationallegalstudies.com/>

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